

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination –2024

Course: B. Tech.

Semester : V

Branch : Electrical and Power/ Electrical and Electronics/ Electrical Engineering

Subject Code & Name: BTEEPLE504B Power Quality Issues

Max Marks: 60

Date: 08/07/2024

Duration: 3 Hrs.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|--|------------|----------|
| A) Explain power quality and explain the reasons for increased concern in power quality | CO1 | 6 |
| B) Discuss the standards of power quality | CO2 | 6 |
| C) With a waveform sketch, explain the terms
(a)Voltage sag
(b)Voltage interruption
(c)Voltage swells | CO2 | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|--|------------|----------|
| A) Explain the sources of sags in power system | CO1 | 6 |
| B) Discuss about the Lightning Phenomenon and also about the functions of lightning arrester | CO2 | 6 |
| C) Explain any two voltage sag mitigation Techniques with necessary circuit diagram and waveform | CO3 | 6 |

Q. 3 Solve Any Two of the following.

- | | | |
|---|------------|----------|
| A) Explain different flicker mitigation techniques. | CO1 | 6 |
| B) Explain the “Ferro Resonance” in detail. | CO2 | 6 |
| C) Explain the various instruments used for power quality measurements. | CO3 | 6 |

Q.4 Solve Any Two of the following.

- | | | |
|---|------------|----------|
| A) Explain the IEEE and IEC standards on harmonics distortion. | CO2 | 6 |
| B) Explain briefly how the phenomena of current distortion affects the voltage distortion under the presence of harmonics | CO3 | 6 |
| C) Write short notes on THD and TDD. | CO4 | 6 |

Q. 5 Solve Any Two of the following.

- | | | |
|--|------------|----------|
| A) Explain the steps involved in power quality monitoring. What are the information's from monitoring site survey? | CO2 | 6 |
| B) Explain different types of monitoring and diagnostic techniques for various power quality problems. | CO3 | 6 |
| C) Briefly discuss the common objectives of power quality monitoring. | CO4 | 6 |

***** End *****